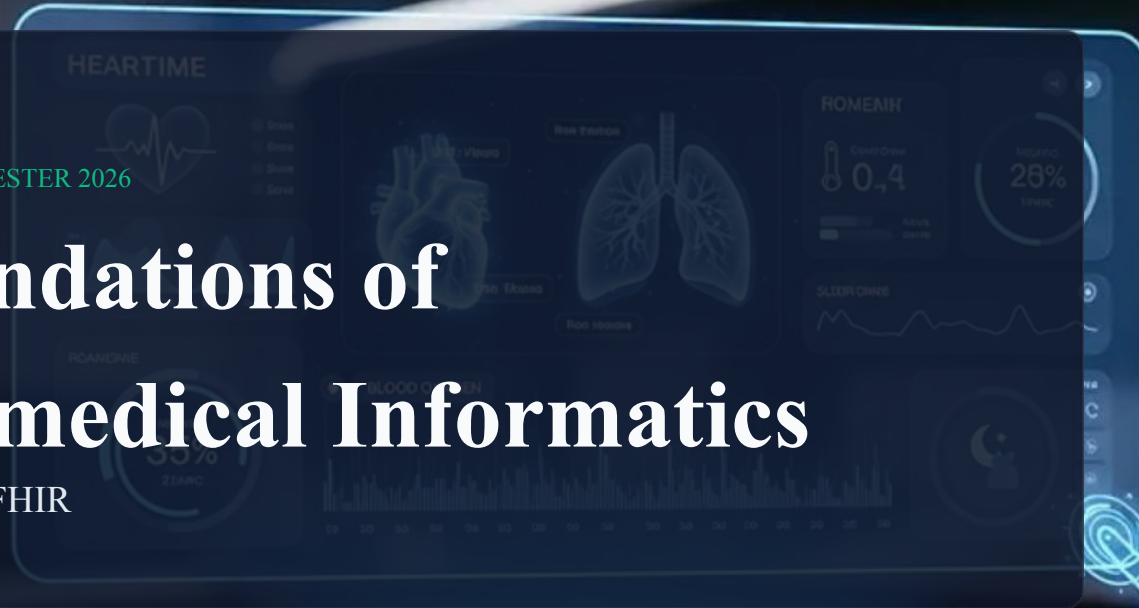


WINTER SEMESTER 2026

Foundations of Biomedical Informatics

Lecture 9: FHIR

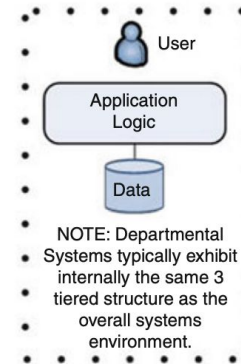
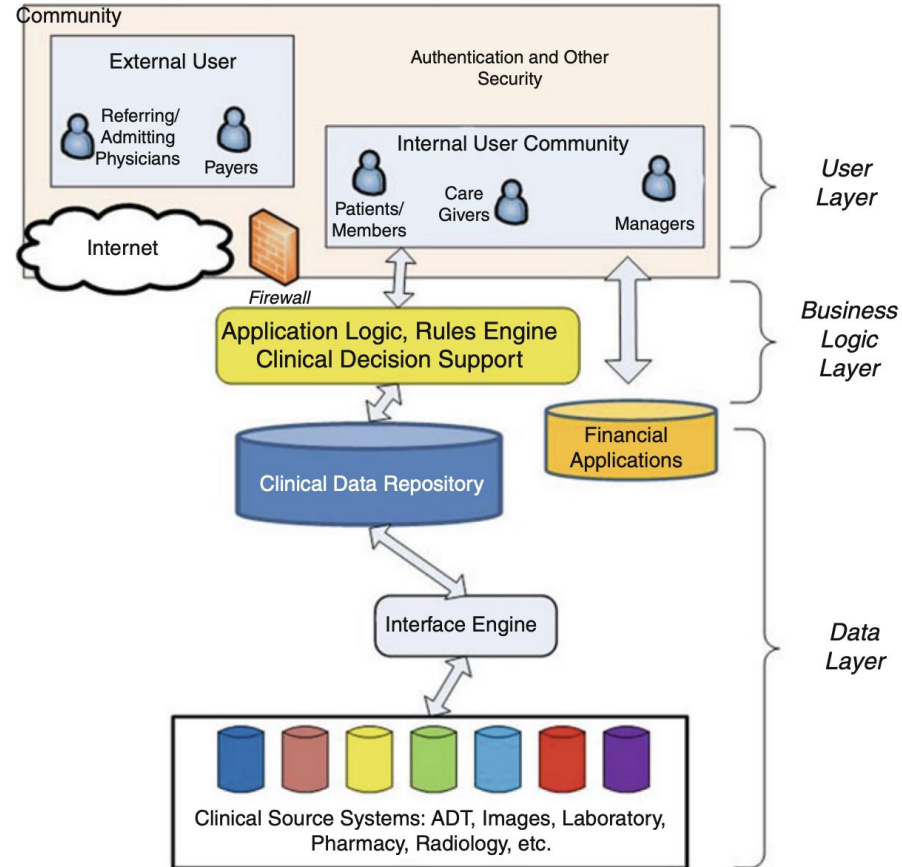


Changing Care Models Around the World

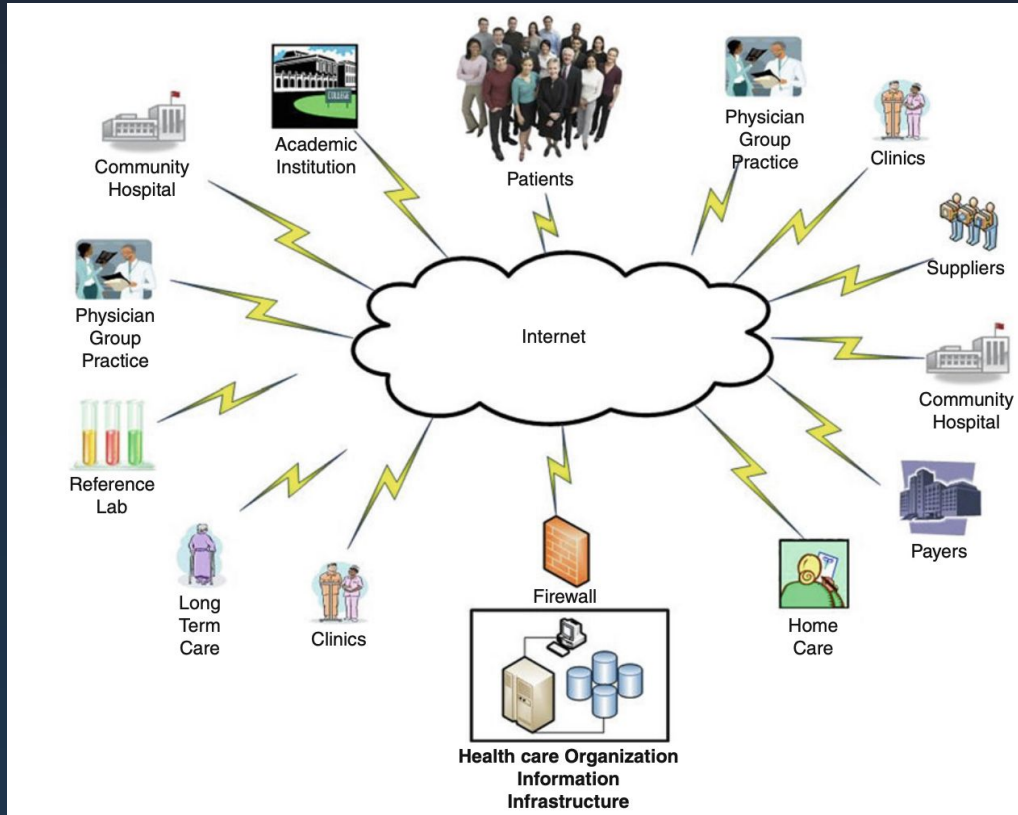
Characteristics	Old care model	New care model
Goal of care	Manage sickness	Manage wellness
Center of delivery system	Hospital	Primary-care providers/ambulatory settings
Focus of care	Episodic acute and chronic care	Population health, primary and preventive care
Driver of care decisions	Specialists	Primary-care providers/patients
Metric of system success	Number of admissions	Number of enrollees
Performance optimization	Optimize individual provider performance	Optimize system-wide performance
Utilization controls	Externally controlled	Internally controlled
Quality measures	Defined as inputs to system	Defined as patient outcomes and satisfaction
Physician role	Autonomous and independent	Member of care team; user of system-wide guidelines of care
Patient role	Passive receiver of care	Active partner in care

Source: Copyright CSC. Reprinted with permission

Design of Hospital Information Systems (HIS)



Chanigng Healthcare Paradigm



Hospital Information Systems (HIS)

Evolution of Computing Systems in Hospitals

ADT/Billing
Systems

Departmental
Systems
(Labs, Radiology, Pharmacy)

Clinical Systems
(Order Entry,
Clinical Documentation)

1960

1970

1980

1990

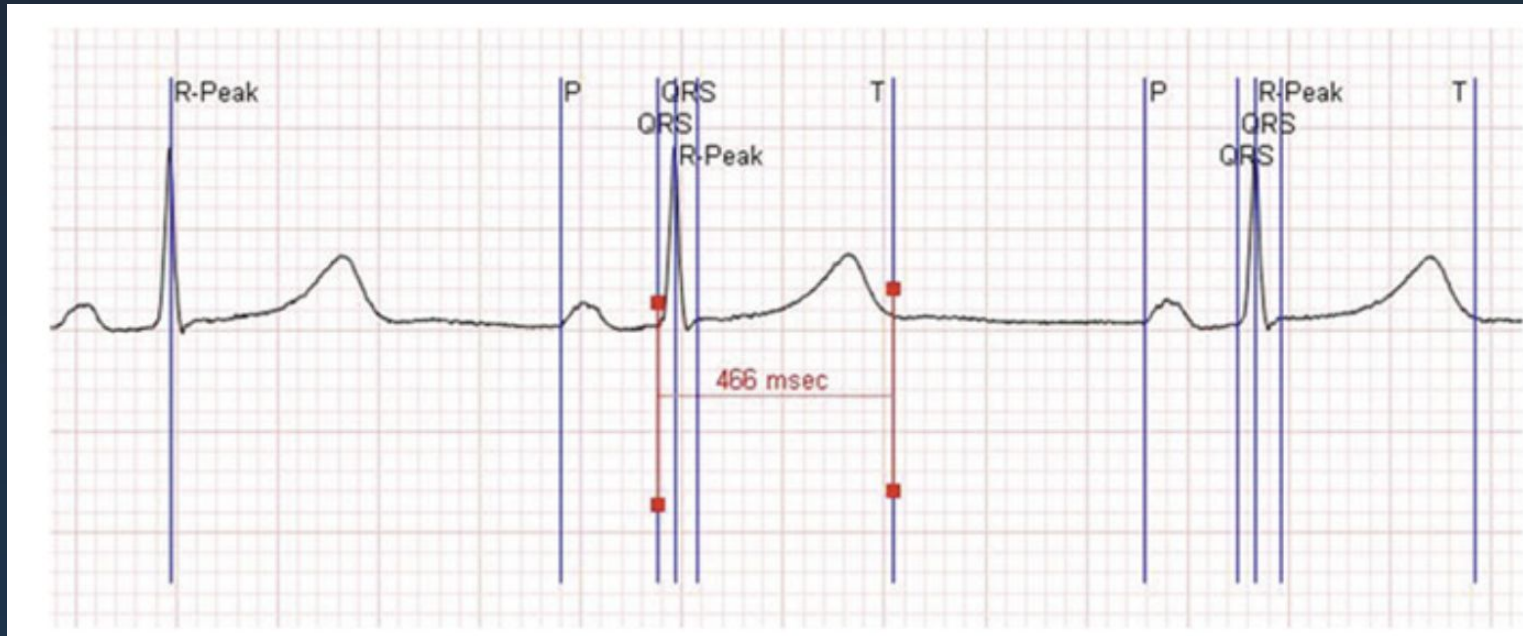
2000

Computers
substitute for
clerical
workers

Productivity
Improvements for
Professional and
Clerical staff in
Departments

Service and Quality
Improvements for
Professionals such
as Physicians and
Nurses

Interoperating with Machine Data



What we want

Report from Lab123457, 15:30 14-Aug-2008, Ref 123456789

Patient: MICKEY MOUSE, DoB: 14-Jan-1962, M

Address: 14 Disney Rd, Disneyland, MM1 9DL

Specimen: Swab, FOOT, Right, Requested By: C987654

Location: 5 N

Patients GP: Dr Smith (G123456)

Organism: STAU

Susceptibility: AMP R

 SXT S

 CIP S

What it looks like at the backend

```
MSH|^~\&||^123457^Labs|||200808141530||ORU^R01|12345678
9|P|2.4
PID|||123456^^^SMH^PI||MOUSE^MICKEY||19620114|M|||14
Disney Rd^Disneyland^^^MM1 9DL
PV1|||5 N||||G123456^DR SMITH
OBR|||54321|666777^CULTURE^LN|||20080802||||||SW^^^FOO
T^RT|C987654
OBX||CE|0^ORG|01|STAU|||||F
OBX||CE|500152^AMP|01|||R|||F
OBX||CE|500155^SXT|01|||S|||F
OBX||CE|500162^CIP|01|||S|||F
```


Technical Interoperability: The Technology Layer

This is the technology layer

- Moves data from system A to system B, neutralizing the effects of distance
- Domain independent
- Does not know or care about the meaning of what is exchanged
- Information theory is the foundation stone of technical interoperability

Health Level 7 International (HL7)

- Is a Standards Development Organization
- Is now the lingua franca to pass health messages across systems
- At least two translations required during information interchange

Internal format ---> Wire Format ---> Internal Format
(computer) (connection) (computer)

Requires the following sequential steps in Object Oriented Software Development

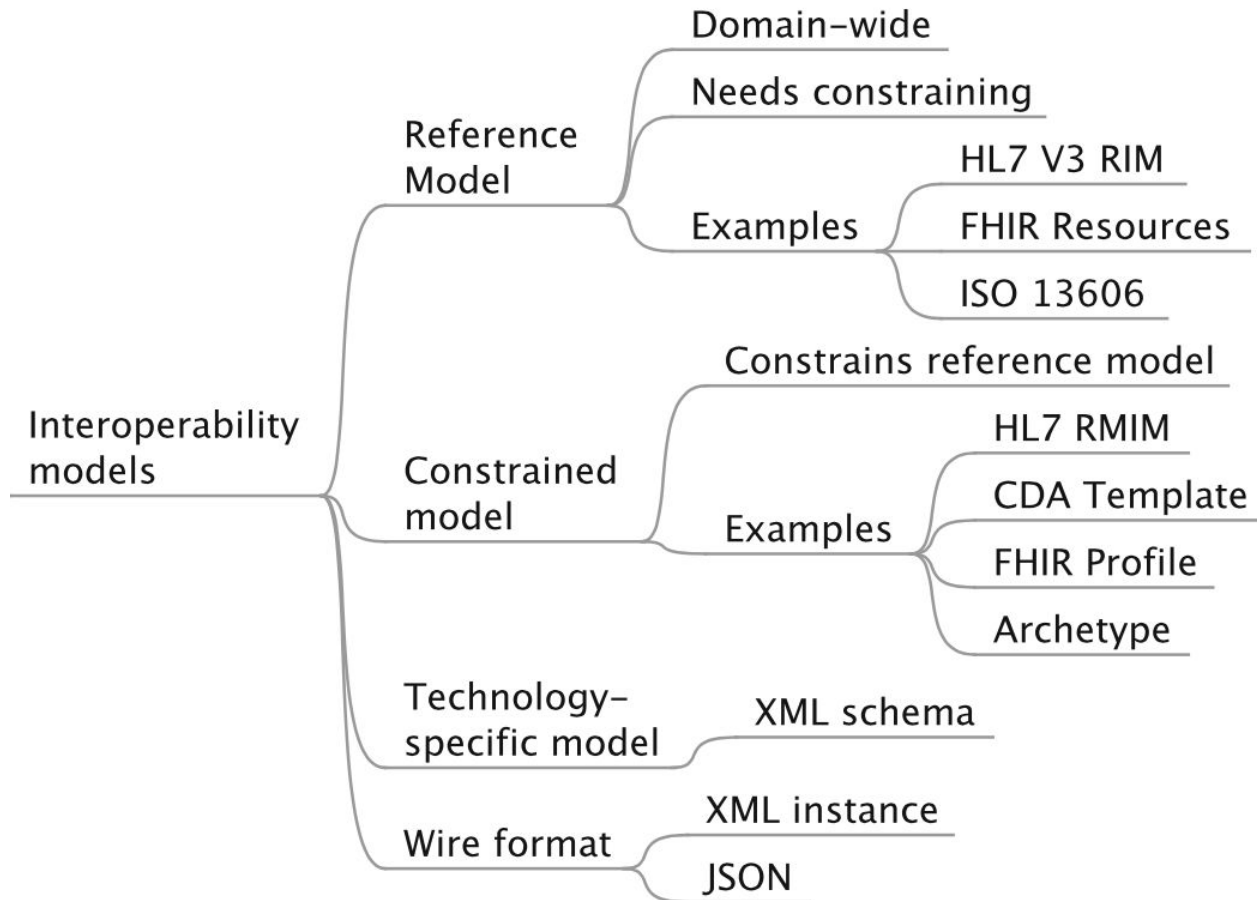
- i. Computational-Independent Model (CIM).
- ii. Platform-Independent Model (PIM) - the conceptual design of a system.
- iii. Platform-Specific Model (PSM)- the implementable design.
- iv. Code, the actual software code, also referred to as wire-format.

Another Example of Syntactic Interoperability: HL7 Standard

```
##### Translate this function from HL7 V2.8 ADT_A01 to HL7 V2.4 ADT_A05
### HL7 V2.8 ADT_A01
```

```
MSH|^~\&|ADT1|GOOD HEALTH HOSPITAL|GHH LAB, INC.|GOOD HEALTH
HOSPITAL|198808181126|SECURITY|ADT^A01^ADT_A01|MSG00001|P|2.8||
EVN|A01|200708181123||
PID|1||PATID1234^5^M11^ADT1^MR^GOOD HEALTH
HOSPITAL~123456789^^^USSSA^SS||EVERYMAN^ADAM^A^III||19610615|M||C|222
2 HOME STREET^^GREENSBORO^NC^27401-1020|GL|(555) 555-2004|(555)555-
2004||S||PATID12345001^2^M10^ADT1^AN^A|444333333|987654^NC|
NK1|1|NUCLEAR^NELDA^W|SPO^SPOUSE|||NK^NEXT OF KIN
PV1|1|I|2000^2012^01|||004777^ATTEND^AARON^A|||SUR|||ADM|A0|

### HL7 V2.4 ADT_A05
```



Segment Definitions, Segments, Reports

```
MSH|^~\&||^123457^Labs|||200808141530||ORU^R01|12345678
9|P|2.4
PID|||123456^^^SMH^PI||MOUSE^MICKEY||19620114|M|||14
Disney Rd^Disneyland^^^MM1 9DL
PV1|||5 N||||G123456^DR SMITH
OBR|||54321|666777^CULTURE^LN|||20080802|||||SW^^^FOO
T^RT|C987654
OBX||CE|0^ORG|01|STAU|||||F
OBX||CE|500152^AMP|01|||||R|||F
OBX||CE|500155^SXT|01|||||S|||F
OBX||CE|500162^CIP|01|||||S|||F
```

Report from Lab123457, 15:30 14-Aug-2008, Ref 123456789

Patient: MICKEY MOUSE, DoB: 14-Jan-1962, M

Address: 14 Disney Rd, Disneyland, MM1 9DL

Specimen: Swab, FOOT, Right, Requested By: C987654

Location: 5 N

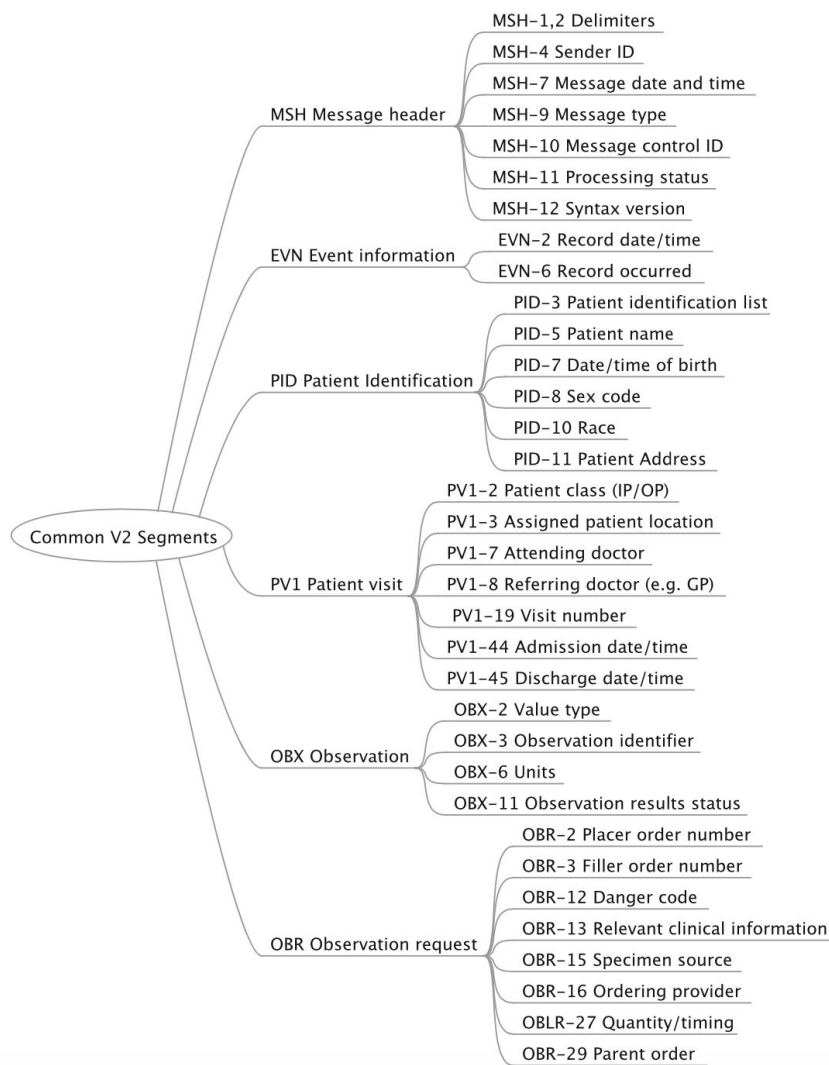
Patients GP: Dr Smith (G123456)

Organism: STAU

Susceptibility: AMP R

SXT S

CIP S



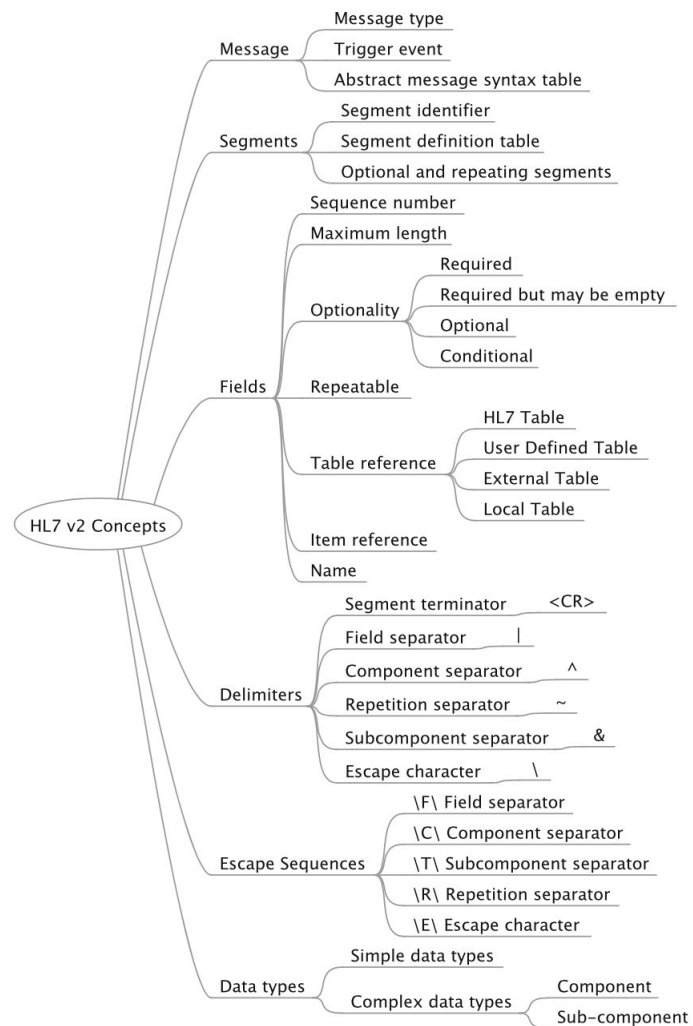
HL7 Messages

Value	Description
ACK	General acknowledgment message
ADT	ADT message
ORM	Order message
ORU	Observation result unsolicited

Value	Description
A01	Admit/visit notification
A02	Transfer a patient
A03	Discharge/end visit
A04	Register a patient

MSH	Message Header
EVN	Event Type
PID	Patient Identification
PV1	Patient Visit

Symbol	Usage
	Field separator
^	Component separator
~	Repetition separator
\	Escape character
&	Subcomponent separator
<CR>	Segment terminator



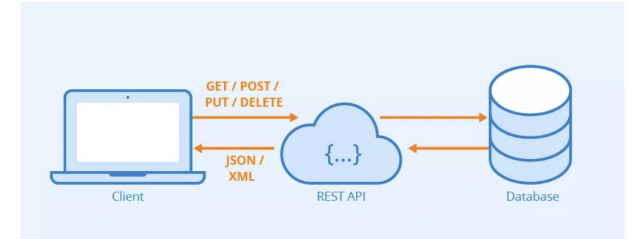
How do you create something like this?

- *public clouds*
- *private clouds*
- *hybrid clouds*

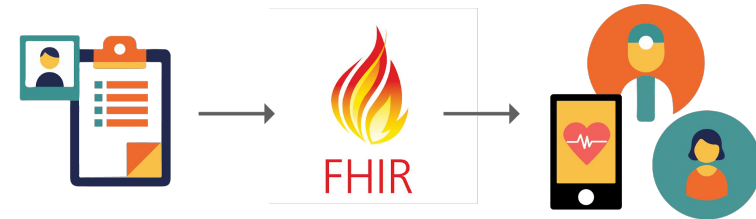


Core Idea: REST

- Representational State Transfer (RESTful) APIs
- Built to copy the web
- Organised around the concept of "Resources"
- All resources have references to other resources, extensions, and a human readable XHTML display
- Augmentation by Messages and Document paradigm also possible
- Open license, a focus on implementation, and a formal maturity process linked to implementation outcome.



Standard REST APIs



FHIR APIs

FHIR Standard

```
```json
{
 "resourceType": "Bundle",
 "type": "message",
 "entry": [
 {
 "resource": {
 "resourceType": "MessageHeader",
 "eventCoding": {
 "system": "http://hl7.org/fhir/message-events",
 "code": "ADT_A01"
 },
 "destination": [
 {
 "endpoint": "http://localhost:8080/fhir/baseDstu3"
 }
],
 "source": {
 "name": "ADT1",
 "software": "GOOD HEALTH HOSPITAL",
 "endpoint": "http://localhost:8080/fhir/baseDstu3"
 }
 }
 }
]
}
```

# Immunization Information Systems (IIS)

## A Public Health Informatics Example

- What is an IIS?
  - Confidential, computerized, population-based systems
  - Collect and consolidate vaccination data
  - Aggregate data from multiple providers
  - Support clinical care and public health decision-making
- 
- Why IIS Matter:
  - Prevent vaccine-preventable diseases
  - Address fragmented records across providers
  - Enable coordinated immunization strategies

# Core Functions of IIS

## Operational Functions:

- Accept electronic & manual immunization records
- Secure 24/7 access to patient immunization history
- Vaccine forecasting & clinical decision support
- Reminder/recall systems
- Coverage reporting by provider, age, geography
- Vaccine inventory management

## CDC Functional Standards:

- HL7-compliant exchange
- Data confidentiality & security
- Accuracy and completeness of records

# Key Informatics Challenges in IIS

## Interdisciplinary Communication:

- Public health experts, IT developers, clinicians, policymakers
- Need shared vocabulary and semantic clarity
- 
- Organizational & Governance Issues:
- Multiple stakeholders
- Transparent decision-making structures
- Policy alignment across jurisdictions

## Funding & Sustainability:

- Limited budgets
- Demonstrating cost-effectiveness & public health value

# System Design & Interoperability

## Interoperability Requirements:

- HL7 messaging standards
- Integration with EHR systems
- Accurate patient matching
- Secure data exchange

## Design Challenges:

- Data acquisition from diverse sources
- Role-based access control
- Consent management
- Interstate data exchange

## Public Health Impact:

- Reduce duplicate vaccinations
- Improve coverage rates
- Support outbreak response and surveillance